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CMPT 466
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Socket Programming Assignment 1

1.

The server sends an ACK when it receives a duplicate message because it needs to notify the client that it has received the client's message. If the server didn't sand an ACK, the client would have no way of knowing if the server received the message and would try to resend the message.

2.

Ping 4 timed out 2 times due to lost messages and once to a lost ACK
Ping 5 timed out once to a lost ACK
Ping 6 timed out once to a lost ACK
Ping 8 timed out once to a lost message.

Lost Messages: 3
Lost ACKs: 3

The client can't tell what caused a timeout. By looking at the server's output, we can tell what caused a timeout. If a message timed out, the message was either lost or the ACK was lost. If the message was lost, the server didn't receive it so when the client retransmits the message the server sees it for the first time and outputs "delivered". If the ACK was lost, the server already saw the message and printed the output "delivered" so when the client retransmits the message, the server outputs "duplicate ignored".

3. A round trip succeeds when neither the message nor the ACK is lost:
P(lost message) = 0.3 P(lost ACK) = 0.2
P(success) = P(message not lost, ACK not lost) = (1 – 0.3) * (1 – 0.2) = 0.7 * 0.8 = 0.56
P(success) = 0.56
Expected sends per message on average:
$P(N = n) = (1 – p)^{n-1}p^n$ where N is the number of sends and n is a possible outcome

$E[N] = \Sigma\, n * P(N = n) = \Sigma\, n(0.44)^{n-1}(0.56)^{n} = 1.786$

My measured result agrees with this since I sent 10 messages and 3 messages timed out once and one message timed out twice.

Client Output

```
Message: Ping 1
response: ACK 1
RTT: 0.001639s

Message: Ping 2
response: ACK 2
RTT: 0.000424s

Message: Ping 3
response: ACK 3
RTT: 0.000294s

Message: Ping 4
Timed out, resending

Message: Ping 4
Timed out, resending

Message: Ping 4
Timed out, resending

Message: Ping 4
response: ACK 4
RTT: 0.001087s

Message: Ping 5
Timed out, resending

Message: Ping 5
response: ACK 5
RTT: 0.001296s
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```
Message: Ping 6
Timed out, resending

Message: Ping 6
response: ACK 6
RTT: 0.000898s

Message: Ping 7
response: ACK 7
RTT: 0.000816s

Message: Ping 8
Timed out, resending

Message: Ping 8
response: ACK 8
RTT: 0.001189s

Message: Ping 9
response: ACK 9
RTT: 0.000739s

Message: Ping 10
response: ACK 10
RTT: 0.000775s

Summary:
Number of messages: 10
Number of packets Sent: 16
Number of packets retransmitted: 6
Total Time: 3.017722s
```

Server Output

